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The Effect of Al Substitution on the Reactivity of Delithiated $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ with Nonaqueous Electrolyte

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The high-temperature reactions between 1 M $LiPF_6$ ethylene carbonate:diethyl carbonate and Al-doped $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ charged to 4.3 V are studied by accelerating rate calorimetry and compared with those of charged $LiNi_{1/3}Mn_{1/3}Co_{1/3}O_2$ and spinel $LiMn_2O_4$. Simultaneous Al substitution for Ni and Mn in $LiNi_{0.5}Mn_{0.5}O_2$ improves the thermal stability. The maximum self-heating rate attained and the specific capacity decrease as the Al content increases. Materials with $z > 0.03$ are less reactive with electrolyte than spinel $LiMn_2O_4$ at all temperatures studied. There is a range of compositions near $z = 0.05$ that show excellent promise as materials which are both safer and more energy dense than spinel $LiMn_2O_4$.
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Lithium nickel manganese oxides have been proposed as alternative positive electrode materials in lithium-ion batteries to replace expensive $LiCoO_2$.[1-6] Of the proposed materials, $LiNi_{0.5}Mn_{0.5}O_2$ is quite attractive due to its high capacity, good capacity retention, economic superiority, and environmental friendliness.[7-10]

Attention to lithium-ion battery safety is needed to avoid battery accidents in the hands of the consumer. Recent recalls of Li-ion batteries suggest that improvements to the response of Li-ion cells to conditions of electrical and mechanical abuse are warranted.[11,12] Inventing and developing electrode materials that are less reactive with the cell electrolyte in their charged state is essential.

Differential scanning calorimetry studies show that charged $LiNi_{0.5}Mn_{0.5}O_2$ does not begin reacting strongly with electrolyte until a significantly higher temperature than charged $LiCoO_2$.[13,14] A consequence of this lower reactivity is that charged Li-ion cells with $LiNi_{0.5}Mn_{0.5}O_2$ positive electrodes can withstand prolonged exposure to higher abusive elevated temperatures ($>130°C$) than can charge $LiCoO_2$ Li-ion cells. However, once charged $LiNi_{0.5}Mn_{0.5}O_2$ begins to react with electrolyte near 260–300°C, the reaction rate accelerates rapidly with temperature. This is shown in Fig. 7 of Ref. 13 and in Fig. 5 of Ref. 14. This rapid reaction rate may lead to thermal runaway in Li-ion cells under internal short-circuit conditions. Barnett et al. show that reducing the maximum rates of heat release caused by the electrode/electrolyte reactions can improve the tolerance of cells to internal short circuits.[15]

Substitution of Al for Ni in $Li[Ni_{1-z}Al_z]O_2$ and substitutions of Mg and Ti for Ni in $Li[Ni_{1-z}Mg_{z/2}Ti_{z/2}]O_2$ have been shown to reduce the rate of exothermic reaction between electrolyte and the charged electrode materials.[16,17] The mechanism for this safety improvement has been studied by Guilmard et al.[18] Al substitution for transition metals in $LiNi_{0.5}Mn_{0.5}O_2$ has been studied recently;[10,19] however, the studies have concentrated on the effect of Al substitution on electrochemical performance. Here, the impact of Al substitution for Ni and Mn in $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ on the reaction rate between the charged electrode material and electrolyte has been studied using accelerating rate calorimetry (ARC) tests. The reactivities of delithiated $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$, $Li[Ni_{1/3}Mn_{1/3}Co_{1/3}]O_2$ (called NMC here), or spinel $LiMn_2O_4$ with electrolyte have been compared.

Experimental

Al-substituted $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ ($z = 0$, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07) samples were synthesized by a coprecipitation-based method. $LiOH{\cdot}H_2O$, $Ni(NO_3)_2{\cdot}6H_2O$, $Mn(NO_3)_2{\cdot}4H_2O$, and $Al(NO_3)_3{\cdot}9H_2O$ (all from Aldrich) were used as starting materials. A 250 mL aqueous solution of metal nitrates (total concentration equal to 0.4 M) and a 250 mL solution of LiOH with corresponding concentration (enough to precipitate all the metal cations) were dripped simultaneously using an electronic metering pump into a stirred flask. The precipitate was rinsed six times with distilled water and dehydrated in air at 80°C overnight. The dried precipitate was mixed with a stoichiometric amount of $LiOH{\cdot}H_2O$ and ground in an automatic grinder. The final products were prepared by heating in air at 500°C for 3 h, then at 1000°C in air for 12 h.

X-ray diffraction (XRD) tests were made using a Siemens D5000 diffractometer. The lattice constants a and c were calculated using least-squares refinements to the positions of at least 10 Bragg peaks. Single-point Brunauer–Emmett–Teller (BET) specific surface area measurements were made using a Micromeritics Flowsorb II 2300 surface area analyzer.

Electrodes were coated on Al foils with a weight-to-weight ratio of 86% of active material, 7% Super-S carbon black, and 7% poly(vinylidene fluoride) binder. The electrodes were dried at 90°C overnight before use. The electrolyte used was 1 M $LiPF_6$ ethylene carbonate (EC):diethyl carbonate (DEC) [1:2 v/v]. 2325 coin-type cells were assembled in a glove box and cycled between 2.5 and 4.3 V using a current of 10 mA/g.

ARC sample-preparation methods are the same as reported in our earlier work.[20-22] Pellet coin cells were made and charged to 4.3 V using the protocol described in Ref. 20. Then, the cells were opened in an argon-filled glove box and the electrode powder was rinsed with dimethyl carbonate four times and dried in the glove box as described in Ref. 21. Charged positive electrode material (94 mg) and 1 M $LiPF_6$ EC:DEC electrolyte (30 mg) was used for the ARC tests. The ARC starting temperature was set to either 70 or 180°C. Exothermic reactions were tracked under adiabatic conditions when the sample self-heating rate (SHR) exceeded 0.03°C/min. Experiments were stopped at 350°C or when the SHR was higher than 20°C/min.

Results and Discussion

All $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ ($z = 0 - 0.07$) samples were single phase by XRD. Table I gives the measured lattice parameters vs z as well as the specific surface area. The lattice constants of $LiNi_{0.5}Mn_{0.5}O_2$ prepared by Ohzuku's group[8] are also shown in Table I for comparison and agree well with our low-z samples. Table I also gives the specific surface areas of cathode materials used for comparison in the ARC experiments, namely, $Li[Ni_{1/3}Mn_{1/3}Co_{1/3}]O_2$ and $Li_{1+x}Mn_{2-x}O_4$.

Figure 1 shows the first charge/discharge curves of $Li/LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ ($z = 0 - 0.14$) cells in the potential range between 2.5 and 4.3 V. Figure 1 shows the substitution of Al for Ni and Mn in $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ affects the electrochemi-

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Table I. Lattice constants and BET surface area vs z in $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$. The surface areas of the comparative commercial materials are also provided.

z value	Lattice constant a (Å)	Lattice constant c (Å)	Surface area (m^2/g)
$z = 0$ from Ref. 8	2.891	14.30	Not available
0.01	2.8859	14.2908	1.171
0.02	2.8851	14.2949	2.137
0.03	2.8804	14.283	2.834
0.04	2.8803	14.291	2.355
0.05	2.8792	14.2958	2.414
0.06	2.8748	14.2775	2.251
0.07	2.8721	14.2805	1.84
$LiNi_{1/3}Mn_{1/3}Co_{1/3}O_2$	2.863	14.24	0.26
$Li_{1+x}Mn_{2-x}O_4$	8.235	Cubic	0.33

cal performance primarily by a reduction in capacity. The initial discharge capacity drops from about 160 to about 120 mAh/g as z increases from 0 to 0.07. Cycling tests to determine capacity retention are still ongoing and will be reported later.

Figure 2 shows the SHR vs temperature for the $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ ($z = 0 - 0.07$) series charged to 4.3 V reacting with 30 mg of 1 M $LiPF_6$ EC:DEC electrolyte with starting temperatures of 70 and 180°C. A 180°C start temperature was included in an attempt to study the thermal stability of $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ under forced heating conditions. Figure 2 shows that when no or little Al is added, the material still reacts dramatically with 1 M $LiPF_6$ EC:DEC when the temperature is higher than 260°C, leading to an extremely rapid temperature rise beyond the maximum heating rate of the ARC. However, when $z \geq 0.02$, the reaction rate is significantly slowed and the SHR never exceeds 20°C/min. For samples with $z \geq 0.05$, the SHR remains below 4°C/min throughout the entire experiment.

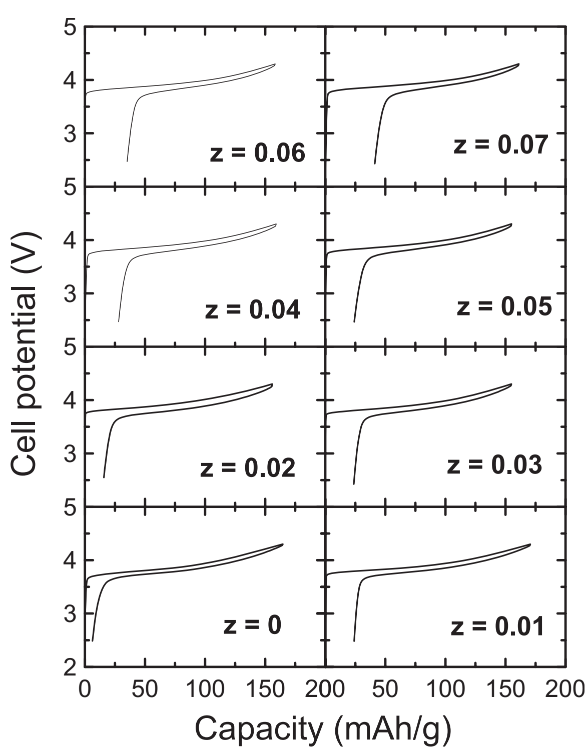

Figure 1. First charge/discharge curves of the $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ series in the potential range between 2.5 and 4.3 V (10 mA/g).

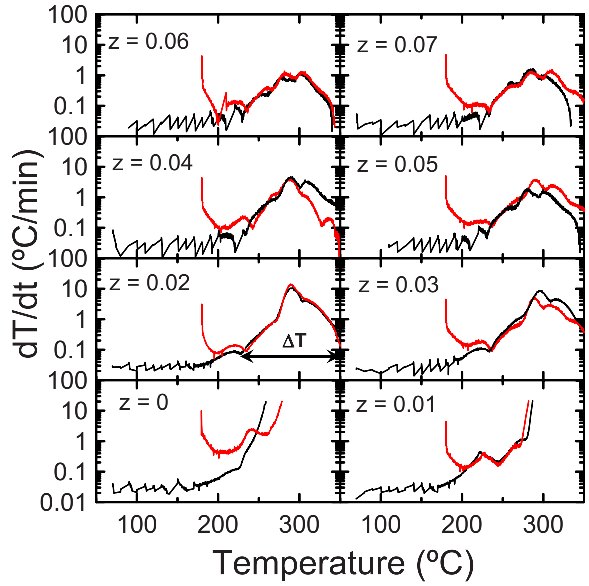

Figure 2. (Color online) SHR vs temperature for delithiated $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ (4.3 V) reacting with 30 mg 1 M $LiPF_6$ EC:DEC with starting temperatures of 70°C (black lines) and 180°C [red lines (online version)].

The temperature rise, ΔT, during the self-sustained exotherm is proportional to the total heat of reaction between the electrode material and the electrolyte. Figure 2 shows that ΔT does not vary much with z for these materials. Therefore, the major effect of Al substitution is a slowing in the kinetics of the reaction.

The addition of Al to $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ causes a reduction in specific capacity. In order to estimate whether the capacity reduction or the presence of Al was responsible for the reduction in thermal reactivity shown in Fig. 2, a series of NMC samples charged to 4.0, 4.1, 4.2, and 4.3 V, respectively, was prepared. These samples had specific capacities of 131, 147, 159, and 168 mAh/g, respectively. It would have been preferable to use commercial $Li[Ni_{1/2}Mn_{1/2}]O_2$ charged to various potentials for this comparison, but we did not have access to such samples.

Figure 3 compares the SHR vs temperature results for these NMC samples with those of the $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ sample with $z = 0.05$ and charged spinel $Li_{1+x}Mn_{2-x}O_4$. Figure 3 suggests that the reduction in reactivity of the Al-substituted samples with z as shown in Fig. 2 is due to the Al content and not to the reduced capacity. In addition, the charged sample with $z = 0.05$ has lower thermal reactivity over the entire temperature range than the charged spinel sample even though it has a much larger BET surface area (Table I). The SHR of the $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ sample with $z = 0.05$ is slightly larger than that of the NMC samples in the range between 180 and 240°C, perhaps due to its much larger (10×) surface area.

Figure 4 shows the maximum SHR from Fig. 2 plotted vs z for the $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ series. The maximum SHR decreases as the Al content increases from greater than 20°C/min to around 1.7°C/min. $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ samples with $z \approx 0.05$ may represent a reasonable trade-off between improved thermal stability and reduced capacity.

Table II gives a comparison of the crystallographic density, the average electrode potential (vs Li), the practical reversible specific capacity, and the volumetric energy density (calculated as the product of the previous three numbers) of some common positive electrode materials compared to $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ ($z = 0.05$). The volumetric energy densities reported in Table II do not take into account the volume of the corresponding negative electrode. Figure

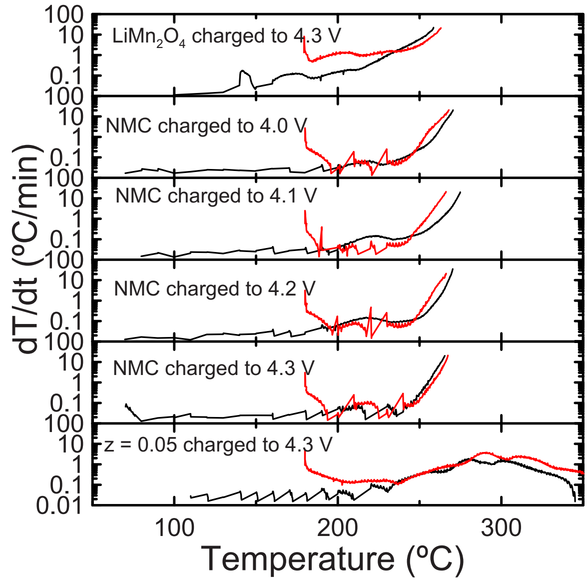

Figure 3. (Color online) SHR vs temperature for charged $Li[Ni_{0.45}Mn_{0.45}Al_{0.1}]O_2$ (4.3 V), NMC charged to 4.0, 4.1, 4.2, and 4.3 V, or spinel $LiMn_2O_4$ reacting with 30 mg 1 M $LiPF_6$ EC:DEC electrolyte with starting temperatures of 70°C (black lines) and 130°C [red lines (online version)].

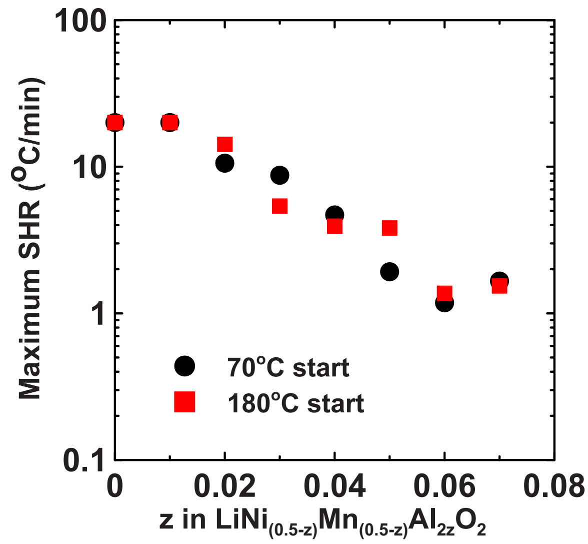

Figure 4. (Color online) Maximum SHR vs Al content, z, for delithiated $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ reacting with 30 mg 1 M $LiPF_6$ EC:DEC with starting temperatures of 70°C (round points) and 180°C (square points).

Table II. Crystallographic density, average electrode potential (vs Li), reversible specific capacity, and volumetric energy density for common positive electrode materials compared to $Li[Ni_{0.45}Mn_{0.45}Al_{0.1}]O_2$.

Material	Crystallographic density (g/cm^3)	Average potential (V)	Reversible specific capacity (Ah/g)	Volumetric energy density (Wh/cm^3)
$LiCoO_2$	5.05	3.9	0.15 (to 4.2 V)	2.95
NMC	4.77	3.7	0.163 (to 4.3 V)	2.87
$Li_{1.1}Mn_{1.9}O_4$	4.18	4.1	0.120	2.06
$LiFePO_4$	3.60	3.44	0.160	1.98
$Li[Ni_{0.45}Mn_{0.45}Al_{0.1}]O_2$	4.51	3.87	0.130 (to 4.3 V)	2.27

3 and Ref. 20 indicate that $Li[Ni_{0.45}Mn_{0.45}Al_{0.1}]O_2$ has better thermal stability than $LiCoO_2$, $LiMn_2O_4$, and NMC. Table II shows that it has better volumetric energy density than both $LiMn_2O_4$ and $LiFePO_4$. Therefore, we believe that Al-substituted $LiNi_{(0.5-z)}Mn_{(0.5-z)}Al_{2z}O_2$ materials may have an important role in large-size Li-ion cells (for plug-in hybrid electric vehicles) where low cost, superior safety, and high energy density are all required.

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